

DDA5001 Machine Learning

Basic Math & Concepts of Learning

Week 1, Part 2

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Basic Mathematics

Concepts of Learning

Basic Notions of Linear Algebra

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- **Vector**. $\mathbf{x} \in \mathbb{R}^n$ is a real-valued n -dimensional **column** vector; i.e.,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad x_i \in \mathbb{R} \ \forall i.$$

- You can regard the vector $\mathbf{x} \in \mathbb{R}^n$ as a point in the n -dimensional **linear space** $\mathbb{R}^n$ (Think of $n = 2$ and $n = 3$).

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- You can regard the vector $\mathbf{x} \in \mathbb{R}^n$ as a point in the n -dimensional **linear space** $\mathbb{R}^n$ (Think of $n = 2$ and $n = 3$).
- **Addition of vectors**. The addition of two vectors is defined by adding corresponding coordinates, i.e.,

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix},$$

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- **Multiplication.** The multiplication of a scalar with a vector is defined by performing multiplication in each coordinate:

$$a \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} ax_1 \\ \vdots \\ ax_n \end{bmatrix},$$

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where $a \in \mathbb{R}$.

- **Commutativity.** $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.
- **Distributive properties** $a(\mathbf{x} + \mathbf{y}) = a\mathbf{x} + a\mathbf{y}$ and $(a + b)\mathbf{x} = a\mathbf{x} + b\mathbf{x}$ for all $a, b \in \mathbb{R}$ and $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.
- **Transpose of vector.** let $\mathbf{x} \in \mathbb{R}^n$. The notation $\mathbf{x}^\top$ means that

$$\mathbf{x}^\top = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}.$$

Basic Notions of Linear Algebra

- **Linear independence.** We say that a finite collection $C = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$ of vectors in $\mathbb{R}^n$ is **linearly dependent** if there exist scalars $a_1, \dots, a_m \in \mathbb{R}$, not all of them are zero, such that

$$\sum_{i=1}^m a_i \mathbf{x}_i = \mathbf{0}.$$

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- **Span.** The set of all linear combinations of $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$ is called the span of $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$, i.e.,
 $\text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\} := \left\{ \sum_{i=1}^m a_i \mathbf{x}_i : \mathbf{a} \in \mathbb{R}^m \right\}$

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- **Basis.** A basis of the n -dimensional space $\mathbb{R}^n$ is a collection of vectors in $\mathbb{R}^n$ that is linearly independent and spans $\mathbb{R}^n$. For example,

$$\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\} \quad \text{and} \quad \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

are bases of $\mathbb{R}^2$.

Basic Notions of Linear Algebra

- **Inner product.** Given two vectors $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{y} \in \mathbb{R}^n$, their inner product is defined as

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i$$

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- **(Euclidean) ℓ_2 -norm.** For vector $\mathbf{x} = [x_1 \ x_2 \ \cdots \ x_n]^\top \in \mathbb{R}^n$,

$$\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^\top \mathbf{x}} = \sqrt{\sum_{i=1}^n x_i^2},$$

which measures the length of $\mathbf{x}$. For simplicity, we often only write $\|\mathbf{x}\|$ to represent $\|\mathbf{x}\|_2$.

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- More generally, a **norm** $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ is a function that satisfies
- $\|\mathbf{x}\| > 0$ for all $\mathbf{x} \neq 0$ and $\|\mathbf{x}\| = 0$ only if $\mathbf{x} = 0$;
 - $\|a\mathbf{x}\| = |a|\|\mathbf{x}\|$ for $\mathbf{x} \in \mathbb{R}^n$ and $a \in \mathbb{R}$;
 - $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ (**triangle inequality**)

Basic Notions of Linear Algebra

- **Hölder p -norm.** We now introduce common norms in $\mathbb{R}^n$ —the Hölder p -norm, $1 \leq p \leq \infty$, which are defined as follows:

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$$

for $1 \leq p < \infty$ and

$$\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

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- **Cauchy-Schwarz inequality.**

$$\mathbf{x}^\top \mathbf{y} \leq \|\mathbf{x}\|_2 \|\mathbf{y}\|_2 \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

Basic Notions of Linear Algebra

- **Matrix.** We use $\mathbb{R}^{m \times n}$ to denote the set of $m \times n$ arrays whose components are from $\mathbb{R}$. We can write a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ as

$$\mathbf{A} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}, \quad a_{i,j} \in \mathbb{R} \quad \forall i, j.$$

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- **Transpose of Matrix.** Given an $m \times n$ matrix $\mathbf{A}$, its transpose $\mathbf{A}^\top$ is defined as the following $n \times m$ matrix:

$$\mathbf{A}^\top = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix},$$

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- **Symmetric matrix.** An $m \times m$ real matrix $\mathbf{A}$ is said to be symmetric if $\mathbf{A} = \mathbf{A}^\top$.

Basic Notions of Linear Algebra

- **Matrix-matrix multiplication.** The matrix-matrix multiplication between $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$ is defined as

$$\mathbb{R}^{m \times p} \ni \mathbf{C} = \mathbf{AB} \quad \text{where} \quad c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Illustration:

$$\begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$$

The matrix-vector multiplication can be viewed as a special case of matrix-matrix multiplication, i.e., with $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^n$, we have

$$\mathbb{R}^m \ni \mathbf{c} = \mathbf{Ab} \quad \text{where} \quad c_i = \sum_{k=1}^n a_{ik} b_k.$$

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- The second one is by **outer product**

$$C = \sum_{k=1}^n \mathbf{a}_k \mathbf{b}_k^{\top},$$

where $\mathbf{a}_k$ and $\mathbf{b}_k^{\top}$ are the k -th column and row of A and B , respectively.

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- The third one is by matrix-vector product

$$\mathbf{c}_j = A \mathbf{b}_j, \quad \forall j = 1, 2, \dots, p.$$

Basic Notions of Linear Algebra

- **Rank.** The rank of a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, denoted by $\text{rank}(\mathbf{A})$, is defined as the number of elements of a maximal linearly independent subset of its columns or rows. Some facts about the rank of a matrix:
- $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top)$;
 - $\text{rank}(\mathbf{A} + \mathbf{B}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B})$;
 - $\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$.

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 - $\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$.
- **Matrix inverse.** An $n \times n$ square matrix $\mathbf{A}$ is said to be invertible if the columns of $\mathbf{A}$ has full-rank. The inverse of the matrix $\mathbf{A}$ is denoted as $\mathbf{A}^{-1}$, and we have

$$\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}.$$

Facts:

- $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$.
- $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$, where $\mathbf{A}, \mathbf{B}$ are square and invertible.

Basic Notions of Linear Algebra

- **Orthogonal matrix.** An $n \times n$ square matrix $\mathbf{A}$ is said to be orthogonal, or orthonormal, if it is a real square matrix whose columns and rows are orthonormal vectors. That is,

$$\mathbf{A}^\top \mathbf{A} = \mathbf{A} \mathbf{A}^\top = \mathbf{I}$$

In another word, for orthogonal matrix $\mathbf{A}$, we have

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- **Positive semi-definite (definite), abbrev. PSD (PD), matrix.** An $n \times n$ real matrix $\mathbf{A}$ is said to be PSD (PD) if $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$ (> 0) for all $\mathbf{x} \in \mathbb{R}^n$ (for all $\mathbf{x} \in \mathbb{R}^n \setminus \{0\}$).

Basic Notions of Multivariate Calculus

Basic Notions of Multivariate Calculus

- **Gradient.** It is a generalization of derivative to multi-dimensional functions. Assume $f(\mathbf{x}) = f(x_1, x_2, \dots, x_n)$ is continuously differentiable. Then, we denote the gradient of f by (an $n \times 1$ vector):

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

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Facts:

- If $f(\mathbf{x}) = \mathbf{c}^\top \mathbf{x}$, then $\nabla f(\mathbf{x}) = \mathbf{c}$.
- If $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{M} \mathbf{x}$ ($\mathbf{M}$ is symmetric), then: $\nabla f(\mathbf{x}) = 2\mathbf{M} \mathbf{x}$.

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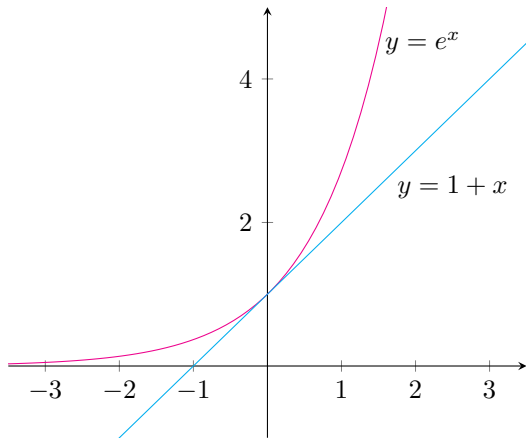
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- **First-order Taylor expansion.** The first-order Taylor expansion yields:

$$f(\mathbf{x} + \mathbf{d}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{d} + o(\|\mathbf{d}\|), \quad \|\mathbf{d}\| \rightarrow 0.$$

Basic Notions of Multivariate Calculus

Illustration of first-order Taylor expansion:



- Approximate the function very well around x .
- Important notion for later first-order algorithm development.

Basic Notions of Probability and Statistics

Basic Notions of Probability and Statistics

- **Expectation.** Suppose X is a random variable, its expectation is denoted as

$$\mathbb{E}[X].$$

Suppose X takes discrete values $x_1, \dots, x_k$ with probability $p_1, \dots, p_k$, then

$$\mathbb{E}[X] = \sum_{i=1}^k p_i x_i.$$

Suppose X takes continuous values in $(-\infty, +\infty)$ with density $p(x)$, then

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} p(x)x dx.$$

- **Variance.** Suppose X is a random variable, its variance is denoted as

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Basic Notions of Probability and Statistics

- ▶ **Random vector.** $\mathbf{X} = [X_1, \dots, X_n]^\top$ is a random vector if each coordinate is a random variable.
- ▶ **Expectation of random vector.** Suppose $\mathbf{X}$ is an n -dimensional random vector, its expectation is denoted as

$$\mathbb{E}[\mathbf{X}] = [\mathbb{E}[X_1], \dots, \mathbb{E}[X_n]]^\top.$$

- ▶ **Covariance matrix.** Suppose $\mathbf{X} = [X_1, \dots, X_n]^\top$ is an n -dimensional random vector, its covariance matrix is $n \times n$ matrix defined as

$$\text{Var}[\mathbf{X}] = \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^\top].$$

Basic Notions of Probability and Statistics

- **Gaussian distribution.** A random variable X is said to follow $\mathcal{N}(\mu, \sigma^2)$ (Gaussian distribution with mean μ and variance σ^2) if its probability density function (PDF) is given by

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

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- **Multivariate Gaussian distribution.** We say the random vector $\mathbf{X} \in \mathbb{R}^d$ follows Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ (assumed to be PD), if its PDF is given by

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

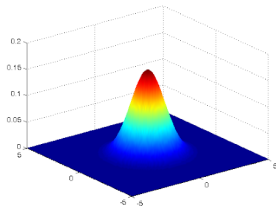
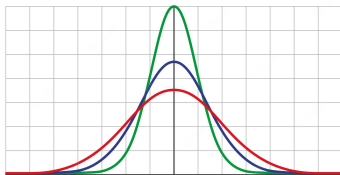
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Basic Notions of Optimization

- **Optimization**. The seek of maximum or minimum. Formally speaking, finding the minimum value of f over $\mathbb{R}^n$ is written as

$$\min_{\boldsymbol{\theta} \in \mathbb{R}^n} f(\boldsymbol{\theta}).$$

Basic Notions of Optimization

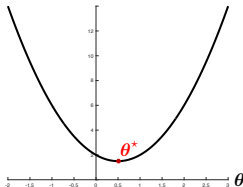
- **Optimization.** The seek of maximum or minimum. Formally speaking, finding the minimum value of f over $\mathbb{R}^n$ is written as

$$\min_{\theta \in \mathbb{R}^n} f(\theta).$$

- **Global minimizer.** Find the point θ^* (called global minimizer/global optimum/optimal solution) that achieves the minimum value of f over $\mathbb{R}^n$

$$\theta^* = \operatorname{argmin}_{\theta \in \mathbb{R}^n} f(\theta).$$

Clearly, $f(\theta^*) = \min_{\theta \in \mathbb{R}^n} f(\theta)$.



Basic Mathematics

Concepts of Learning

Components of Supervised Learning: Motivation from An Example

Learning Example: Credit Approval

Task: Learning to predict if one applicant should be approved a credit card.

Applicant's info.:

age	25
gender	male
salary	100000 RMB
citizenship	CN
years in job	2 year
⋮	⋮

Question: should we approve credit card to the applicant?

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Question: should we approve credit card to the applicant?

How to automate such a task by using machine learning methods?

Data: Samples

- Collect a series of **historical data**

	age	gender	salary	citizenship	years in job
Applicant 1	2.5	1	10	3	1
Applicant 2	2.8	0	8	6	5
Applicant 3	1.6	0	0	4	0
Applicant 4	2.3	1	8	2	4
Applicant 5	3	0	4	2	1

We have the **data matrix** (**feature matrix**) $\mathbf{X}$ as

$$\mathbf{X} = \begin{bmatrix} 2.5 & 1 & 10 & 3 & 1 \\ 2.8 & 0 & 8 & 6 & 5 \\ 1.6 & 0 & 0 & 4 & 0 \\ 2.3 & 1 & 8 & 2 & 4 \\ 3 & 0 & 4 & 2 & 1 \end{bmatrix}$$

- Each row $\mathbf{x}_i^\top$ is called a **sample**, representing i -th applicant's data
- Each column is called a **feature**, representing all the applicant's behavior about the j -th feature.

Data: Labels

- Collect the corresponding label

	age	gender	salary	citizenship	years in job	approve
App. 1	2.5	1	10	3	1	+1
App. 2	2.8	0	8	6	5	+1
App. 3	1.6	0	0	4	0	-1
App. 4	2.3	1	8	2	4	+1
App. 5	3	0	4	2	1	-1

We have the label y as

$$y = \begin{bmatrix} +1 \\ +1 \\ -1 \\ +1 \\ -1 \end{bmatrix}$$

- Each y_i represents the label of the i -th applicant.
- Label implies supervision.

Supervised Learning: Hypothesis/Model

- ▶ We have an underlying and unknown hypothesis/model $g \in \mathcal{H}$

$$g : \mathcal{X} \mapsto \mathcal{Y}$$

where $\mathcal{X}$ is the input space (set of all possible inputs), while $\mathcal{Y}$ is the output space (label space).

In our example, g is the target function that maps x_i to y_i .

- ▶ Learn a model f from the hypothesis/model space $\mathcal{H}$ based on the training dataset $\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)\}$.

Ideally, f should fully capture the patterns in data, i.e., it approximates well the target function g

$$f \approx g.$$

- ▶ The hypothesis space $\mathcal{H}$ is one of the hardest parts to be pre-determined in a learning process. One typical instance of $\mathcal{H}$ is the set of all possible linear fit to the data (results in linear models), while another popular choice is nonlinear model (e.g., neural networks).

Supervised Learning: Hypothesis/Model

Parametrization:

$f = f_{\theta} \in \mathcal{H}$ is often parameterized by the parameters θ

Example:

- ▶ In linear regression, $f_{\theta}(x) = \theta^{\top} x$ is all possible linear fits and θ is the parameters of the model. A specific θ determines a specific model.
- ▶ In deep learning, f_{θ} is the neural network and θ represents weights (network parameters), respectively.

Supervised Learning: Hypothesis/Model

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Two main categories of hypothesis space $\mathcal{H}$:

- ▶ **Linear**
 - ▶ Linear regression
 - ▶ Linear classification
- ▶ **Nonlinear**
 - ▶ Neural networks

Supervised Learning: Learning Problem and Algorithm

- ▶ Given training dataset $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)$.
- ▶ Choose the hypothesis f_{θ} .
- ▶ Choose the **loss function** $\ell : \mathbb{R} \rightarrow \mathbb{R}$.
- ▶ **Learning/optimization problem**

$$\hat{\theta} = \operatorname{argmin}_{\theta \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n \ell(f_{\theta}(\mathbf{x}_i), y_i) \quad (\mathbf{P})$$

$\rightsquigarrow$ **Optimization algorithm** $\mathcal{A}$ is designed to solve $(\mathbf{P})$.

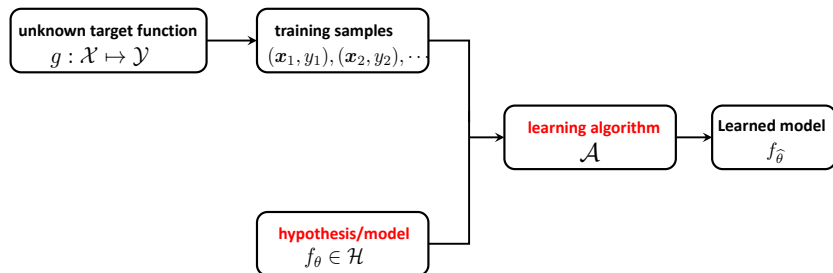
$\rightsquigarrow$ After learning to obtain $\hat{\theta}$, we get the **learned model** $f_{\hat{\theta}}$. Then, one can use the learned $f_{\hat{\theta}}$ to do prediction.

Supervised Learning: Components

Formalization:

- ▶ Target function $g : \mathcal{X} \rightarrow \mathcal{Y}$ (underlying credit approval model)
- ▶ Training dataset: $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)$ (historical records)
- ▶ Hypothesis space $\mathcal{H}$ (learning scope to approximate g)
- ▶ Hypothesis/model: f_{θ} (model to be determined)
- ▶ Optimization algorithm: $\mathcal{A}$ (learning the model from data)

Supervised Learning: High-level View



Hopefully,

$$f_{\hat{\theta}} \approx g$$

Predict/decision: When a new sample data (test data) x comes, the label is predicted as:

$$y \leftarrow f_{\hat{\theta}}(x).$$

⇒ Next lecture: Linear classification and linear regression.